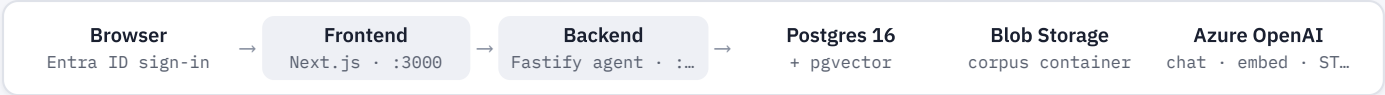


Infrastructure Setup — One-Pager

Target Azure · UAE North · Profile Small pilot (4–5 users) · Reviewed 29 Sep 2026

What to provision to run this app for real. Grounded in the actual code, not a generic checklist — today it runs against public OpenAI with zero Azure resources provisioned. Target: Azure OpenAI only, PTU, UAE North.

SHAPE OF THE SYSTEM



PROVISION THIS

COMPONENT	STATUS	SETUP	ENV VAR
SSO — Entra ID	REQUIRED	App registration, platform SPA . Redirect <code><frontend>/auth/microsoft/callback</code> . Scopes <code>openid profile email User.Read</code> . Client/tenant IDs are baked in at build time — a wrong value means a rebuild, not a config edit. Backend validates no tokens; this covers the browser only.	NEXT_PUBLIC_AZURE_CLIENT_ID NEXT_PUBLIC_AZURE_TENANT_ID
Network / private endpoints	REQUIRED	Private endpoint on Postgres, Storage, Azure OpenAI . Backend Container App joins the VNet; ingress internal-only — it has no auth of its own. Any gateway in front of the chat stream must disable response buffering (SSE) or streaming breaks silently.	CORS_ORIGINS
Database	REQUIRED	Postgres Flexible Server 16. Allow-list vector under <code>azure.extensions</code> before first boot — the app runs <code>CREATE EXTENSION</code> itself. Always <code>sslmode=require</code> .	DATABASE_URL
Storage	REQUIRED	Blob container <code>undersecretary-companion-corpus</code> (already the code default). Backend's managed identity needs Storage Blob Data Contributor at account scope.	STORAGE_BACKEND=blob AZURE_STORAGE_ACCOUNT_URL
AI / PTU	REQUIRED — PTU	Azure OpenAI only, no public OpenAI fallback . Provisioned Throughput (PTU) deployment in UAE North , per model (<code>gpt-5</code> , <code>text-embedding-3-large</code> , TTS/STT) — reserved capacity, billed hourly regardless of use. Confirm each model's PTU availability in UAE North first ; request quota early, it is usually the longest lead time in this list.	OPENAI_API_BASE
Secrets	RECOMMENDED	Key Vault for <code>OPENAI_API_KEY</code> , <code>PGPASSWORD</code> , session secrets — referenced by the Container Apps, never inlined into app settings.	—
Compute	REQUIRED	Azure Container Apps — one frontend, one backend, fixed at 1 replica each, no autoscale . Sized for 4–5 named users, not public traffic; default consumption sizing (0.5–1 vCPU / 1–2 GiB) covers the boot-time embedding warm-up. Built-in ingress is SSE-safe by default, which App Gateway is not without extra care.	PORT=8000
News feed	STUB TODAY	The News page reads a static bundled catalog (<code>newsArticles.json</code>) — <code>fetchBloombergItems</code> in the backend is a marked stub, no live call exists yet. A real provider needs its own API key in Key Vault, an explicit egress rule out of the otherwise internal-only backend network, and a refresh job (Container Apps job or timer-triggered Function) to pull new articles in.	— (future: news provider key)

PROVISIONING ORDER

1. Request Azure OpenAI PTU quota in UAE North (start first — longest lead time) → 2. VNet + subnets → 3. Key Vault → 4. Postgres (private endpoint) → 5. Storage account + container → 6. Azure OpenAI PTU deployments (once approved) → 7. Entra ID app registration → 8. Backend Container App (internal ingress) → 9. Frontend Container App (public ingress) → 10. add the final redirect URI back into the app registration.

! Before reusing `deploy/` or `docs/AZURE_DEPLOYMENT_DESIGN.md` : both were inherited from a sister project ("OPANRA" / Petroleum Companion) and still carry its names — Key Vault `dof-kv-n-oap-*` , ACR repo `opanra/backend` , container `opanra-ai-petroleum-companion` , database `petroleum_companion` . Rename to this app's own resources first.